FORMAL LOCAL STRUCTURE OF A FLAT FAMILY AT A SPLIT NODE OF THE FIBRE: COMPLETED LOCAL RING IS O_{S,S}^WEDGE[[X, Y]]/(XY - H)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the localization and completion setup, the split-node presentation of B/mB, the full correction step with its explicit formulas for u', v', h' and delta', the iteration with its Cauchy estimates, and the closing surjectivity and mod-injectivity arguments are all retained verbatim.

Editorial note. The label lemma-formal-local-structure-nodal-family names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Split nodes of the central fibre. Same chapter, Lemma lemma-split-node and Definition definition-split-node. Supplies the mod-m normal form: (A/m)[[x, y]] -> B/mB with x, y mapping to u-bar, v-bar is surjective with kernel (xy).
- Flatness of completed local rings along the family. More on Algebra, Lemma more-algebra-lemma-flat-completion; Algebra, Lemmas algebra-lemma-completion-generalities and algebra-lemma-mod-injective. Gives flatness of A -> B for the completions, surjectivity from the mod-m surjection, and the lifting of mod-m isomorphisms of flat complete algebras.

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at proof line(s) 5229; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00ME: sources/stacks-chow/tags.txt, source lines 820–820. Prerequisite pointer invoked at proof line(s) 5230; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBW: sources/stacks-curves/pinned-original-source.tex, source lines 4585–4585. Prerequisite pointer invoked at proof line(s) 5164; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0C4G: sources/stacks-chow/tags.txt, source lines 15226–15226. Prerequisite pointer invoked at proof line(s) 5160; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : T \to S$ be a morphism of schemes. Let $t \in T$ with image $s \in S$. Assume*

(1) *f is flat at t,*
(2) *$\mathcal{O}_{S,s}$ is Noetherian,*
(3) *f is locally of finite type,*
(4) *t is a split node of the fibre T_s.*

Then there exists an $h \in \mathfrak{m}_s^\wedge$ and an isomorphism

$$\mathcal{O}_{T,t}^\wedge \cong \mathcal{O}_{S,s}^\wedge[[x, y]]/(xy - h)$$

of $\mathcal{O}_{S,s}^\wedge$-algebras.

Proof. We replace S by $\mathrm{Spec}(\mathcal{O}_{S,s})$ and T by the base change to $\mathrm{Spec}(\mathcal{O}_{S,s})$. Then T is locally Noetherian and hence $\mathcal{O}_{T,t}$ is Noetherian. Set $A = \mathcal{O}_{S,s}^\wedge$, $\mathfrak{m} = \mathfrak{m}_A$, and $B = \mathcal{O}_{T,t}^\wedge$. By More on Algebra, Lemma 0C4G we see that $A \to B$ is flat. Since $\mathcal{O}_{T,t}/\mathfrak{m}_s\mathcal{O}_{T,t} = \mathcal{O}_{T_s,t}$ we see that $B/\mathfrak{m}B = \mathcal{O}_{T_s,t}^\wedge$. By assumption (4) and Lemma 0CBW we conclude there exist $\overline{u}, \overline{v} \in B/\mathfrak{m}B$ such that the map

$$(A/\mathfrak{m})[[x, y]] \longrightarrow B/\mathfrak{m}B, \quad x \longmapsto \overline{u}, x \longmapsto \overline{v}$$

is surjective with kernel (xy).

Assume we have $n \geq 1$ and $u, v \in B$ mapping to $\overline{u}, \overline{v}$ such that

$$uv = h + \delta$$

for some $h \in A$ and $\delta \in \mathfrak{m}^n B$. We claim that there exist $u', v' \in B$ with $u - u', v - v' \in \mathfrak{m}^n B$ such that

$$u'v' = h' + \delta'$$

for some $h' \in A$ and $\delta' \in \mathfrak{m}^{n+1}B$. To see this, write $\delta = \sum f_i b_i$ with $f_i \in \mathfrak{m}^n$ and $b_i \in B$. Then write $b_i = a_i + ub_{i,1} + vb_{i,2} + \delta_i$ with $a_i \in A$, $b_{i,1}, b_{i,2} \in B$ and $\delta_i \in \mathfrak{m}B$. This is possible because the residue field of B agrees with the residue field of A and the images of u and v in $B/\mathfrak{m}B$ generate the maximal ideal. Then we set

$$u' = u - \sum b_{i,2} f_i, \quad v' = v - \sum b_{i,1} f_i$$

and we obtain

$$u'v' = h + \delta - \sum (b_{i,1}u + b_{i,2}v) f_i + \sum c_{ij} f_i f_j = h + \sum a_i f_i + \sum f_i \delta_i + \sum c_{ij} f_i f_j$$

for some $c_{i,j} \in B$. Thus we get a formula as above with $h' = h + \sum a_i f_i$ and $\delta' = \sum f_i \delta_i + \sum c_{ij} f_i f_j$.

Arguing by induction and starting with any lifts $u_1, v_1 \in B$ of $\overline{u}, \overline{v}$ the result of the previous paragraph shows that we find a sequence of elements $u_n, v_n \in B$ and $h_n \in A$ such that $u_n - u_{n+1} \in \mathfrak{m}^n B$, $v_n - v_{n+1} \in \mathfrak{m}^n B$, $h_n - h_{n+1} \in \mathfrak{m}^n$, and such that $u_n v_n - h_n \in \mathfrak{m}^n B$. Since A and B are complete we can set $u_\infty = \lim u_n$, $v_\infty = \lim v_n$, and $h_\infty = \lim h_n$, and then we obtain $u_\infty v_\infty = h_\infty$ in B. Thus we have an A-algebra map

$$A[[x, y]]/(xy - h_\infty) \longrightarrow B$$

sending x to u_∞ and v to v_∞. This is a map of flat A-algebras which is an isomorphism after dividing by $\mathfrak{m}$. It is surjective modulo $\mathfrak{m}$ and hence surjective by completeness and Algebra, Lemma 0315. Then we can apply Algebra, Lemma 00ME to conclude it is an isomorphism. □